Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit. Questions refer to the nonindented paragraph above unless otherwise stated.

Question permutation table:

Version		1	2	3	4
Question					
1	→	8	7	1	10
2*	→	9	8	2	11
3	→	4	12	3	12
4*	→	5	13	4	13
5*	→	6	14	5	14
6*	→	7	15	6	15
7	→	15	4	10	2
8	→	14	9	14	8
9	→	1	10	11	3
10*	→	2	11	12	4
11	→	3	6	13	1
12	→	13	5	15	9
13	→	10	1	7	5
14*	→	11	2	8	6
15*	→	12	3	9	7

*: Question follows previous

The following table shows the responses from a sample of 680 people in the General Social Survey to the question, "Do you sometimes drink more than you think you should?"

Gender	Yes	No	Total
Male	151	177	328
Female	92	260	352
Total	243	437	680

1. What proportion of males responded that they drink more than they should?

   (a) 151/243
   (b) 151/328
   (c) 243/680
   (d) 151/680

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

2. Does there appear to be an association between gender and whether a person responds that they drink more than they should?

   (a) Yes, since a different number of people of each gender are in the sample.
   (b) Yes, since the percent that responded that they drink more than they should differs from 50
   (c) Yes, since the proportion who responded they drink more than they should differs between males and females.
   (d) No, there does not appear to be an association between these two variables.

   Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)

A study published in 2007 by Christopher Johnson, professor of music education and music therapy at the University of Kansas, revealed that students in elementary schools who took private music lessons scored around 20 percent higher in math scores on standardized tests, compared to students who did not take private music lessons. Use this to answer the following four questions:

3. Is this a randomized experiment or an observational study?

   (a) Randomized experiment, since researchers gave students standardized tests.
   (b) Observational study, since whether the student took private music lessons was not randomly assigned.
   (c) Observational study, since math scores were not randomly assigned.
   (d) Observational study, since this was not a random sample.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(b)	(b)	(c)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(d)	(c)	(d)	(c)

4. What is the explanatory variable?

   (a) Music therapy
   (b) Whether a student took private music lessons
   (c) Math score on standardized test
   (d) Grade level

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(c)	(b)	(b)	(c)
(d)	→	(b)	(d)	(a)	(b)

5. What is the response variable?

   (a) Music therapy
   (b) Whether a student took private music lessons
   (c) Math score on standardized test
   (d) Grade level

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(c)	(b)	(d)	(d)
(c)	→	(d)	(d)	(a)	(b)
(d)	→	(b)	(c)	(b)	(c)

6. Which of the following could be a potential confounding variable?

   (a) Household income
   (b) Socio-economic status
   (c) Whether the parents hired a math tutor
   (d) All of these

Correct answers: (d)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(d)	(c)
(b)	→	(d)	(c)	(a)	(b)
(c)	→	(b)	(d)	(c)	(d)
(d)	→	(a)	(b)	(b)	(a)

Researchers randomly assigned patients with severe acne to one of two treatments: the standard topical antibiotic (the control) or a new antibiotic taken orally (the treatment). Six months later an evaluation of skin problems was made by a doctor who did not know which medication subjects had been given.

7. What is the primary advantage of random assignment of the treatment?

   (a) Random assignment ensures that the study participants are representative of the larger population.
   (b) Random assignment allows inference back to a population of patients with acne.
   (c) Random assignment allows the researchers to infer that any difference in outcome was due to the treatments.
   (d) Random assignment ensures that the subject doesn't know which treatment they are getting.

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(a)	(b)
(b)	→	(c)	(a)	(b)	(a)
(c)	→	(d)	(c)	(d)	(d)
(d)	→	(b)	(b)	(c)	(c)

Recently, a Montana newspaper asked people to go to the newspaper's website and respond yes or no to the question: "Should federal protections on grizzly bears be lifted?" Of 933 opinions, 58.1% said "Yes", and 41.9% said either "No" or "Uncertain".

8. Is it valid to conclude from the poll that over half of all Montana residents favor lifting the protections?

   (a) Yes, because 58.1% is far enough above 50% to conclude that the true proportion is more than one-half.
   (b) Yes, because the sample size is large enough to represent all 75,000 Montana residents.
   (c) No, because the sample is not likely to be representative of the population.
   (d) No, because the sample is too small given the population of 75,000 Montana residents.

Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(a)	(d)
(b)	→	(d)	(c)	(d)	(b)
(c)	→	(c)	(a)	(c)	(c)
(d)	→	(a)	(d)	(b)	(a)

In a study on treatments for anorexia, researchers recruited 30 teenage girls with anorexia. The girls were ranked by weight prior to the study. Of the heaviest three girls, one was randomly assigned to a "family" therapy, one to a "cognitive/behavioral" therapy, and one to a control group. The same was done for the next three heaviest girls, down to the three lightest girls. After five weeks of treatment, researchers weighed the participants. Researchers would like to determine which, if any, of the treatments are most effective in treating anorexia. Use this scenario to answer questions 9 and 10.

9. If the researchers found that there were statistically significant differences in average weight across the three treatments, would the researchers be justified in concluding that the treatments caused the changes in weights?

   (a) No, since the researchers do not know which treatment was most effective.
   (b) No, since weight was not randomly assigned.
   (c) Yes, since the therapy group was randomly assigned.
   (d) Yes, since this was an observational study.

   Correct answers: (c)
   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(c)	(d)
(b)	→	(b)	(c)	(d)	(b)
(c)	→	(d)	(b)	(b)	(a)
(d)	→	(a)	(d)	(a)	(c)

10. What type of study design did the researchers use?

    (a) Completely randomized design
    (b) Matched pairs design
    (c) Blocking study design
    (d) Observational study

    Correct answers: (c)
    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(b)	(d)	(b)
(b)	→	(d)	(a)	(b)	(d)
(c)	→	(b)	(c)	(a)	(a)
(d)	→	(a)	(d)	(c)	(c)

Sickle-cell disease is a painful disorder of the red blood cells. To investigate whether drug A can reduce the pain associated with sickle-cell disease, a study by the National Institutes of Health randomly assigned 150 sickle-cell sufferers to receive the drug. Placebos were given to another 150 sickle-cell sufferers. Great care was used to ensure that the 300 participants did not know if their pill contained the drug. At the end of the treatment period, the researchers counted the number of episodes of pain reported by each subject.

11. Which of the following principles was NOT used in this study?

    (a) Control group
    (b) Blinding
    (c) Randomization
    (d) Blocking

    Correct answers: (d)
    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(a)	(c)	(a)
(b)	→	(b)	(c)	(b)	(d)
(c)	→	(c)	(b)	(d)	(c)
(d)	→	(a)	(d)	(a)	(b)

Whirling disease is a deadly disease that affects trout in Montana rivers. In a follow-up to a 2006 study conducted by the Montana Department of Fish, Wildlife and Parks (FWP), researchers sought to determine if the proportion of trout afflicted by whirling disease in the Gallatin river differs between rainbow trout and brown trout. To test this theory, researchers collected a representative sample of 527 rainbow trout and 459 brown trout. Of the 527 rainbow trout collected, 120 had developed whirling disease; of the 459 brown trout collected, 74 had developed whirling disease.

12. Does whether a trout develops whirling disease appear to be associated with the species of trout (rainbow or brown)?

    (a) No, since approximately the same number of trout in each species were used.
    (b) Yes, since the number of trout who developed whirling disease differed from the number of trout who did not develop whirling disease.
    (c) Yes, since the conditional proportion of trout that developed the disease differed between rainbow and brown trout.
    (d) Yes, since the proportion of trout that developed the disease was not equal to 0.5.

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(a)	(d)	(b)
(b)	→	(a)	(b)	(c)	(a)
(c)	→	(d)	(d)	(a)	(d)
(d)	→	(b)	(c)	(b)	(c)

The Women's Health Study included 39,876 female health professionals aged 45 years and older who were followed for an average of 10 years (Ridker et al., 2005). At the beginning of the study, participants were randomly assigned to take either a daily low-dose aspirin or a daily placebo pill for the duration of the study. At the end of the study, researchers measured the number of participants that suffered from a heart attack during the study. Data are summarized in the following table.

	Low-dose Aspirin	**Placebo**	**Total**
Heart Attack	198	193	391
No Heart Attack	19,736	19,749	39,485
Total	19,934	19,942	39,876

Use this scenario to answer questions 13, 14, and 15.

13. What type of plot could be used to plot these data?

    (a) Scatterplot
    (b) Segmented bar chart
    (c) Dotplot
    (d) Histogram

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(b)	(d)	(b)
(b)	→	(b)	(a)	(b)	(a)
(c)	→	(c)	(d)	(a)	(c)
(d)	→	(a)	(c)	(c)	(d)

14. Calculate the relative risk of heart attack for the low-dose aspirin group compared to the placebo group.

    (a) 1.026
    (b) 0.974
    (c) 0.00993
    (d) 0.000255

    Correct answers: (a)

    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(c)	(d)	(b)
(b)	→	(d)	(a)	(a)	(c)
(c)	→	(a)	(d)	(c)	(a)
(d)	→	(b)	(b)	(b)	(d)

15. Do the data suggest an association exists between whether a woman takes low-dose Aspirin or placebo and if the woman suffers from a heart attack in this sample?

    (a) No, since approximately the same proportion of women suffered from a heart attack in each group.
    (b) Yes, women who took low-dose Aspirin were less likely to suffer from a heart attack.
    (c) Yes, women who took low-dose Aspirin were more likely to suffer from a heart attack.
    (d) We cannot determine whether an association exists without a plot.

    Correct answers: (a)

    Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(d)	(c)	(a)	(d)
(b)	→	(c)	(a)	(d)	(c)
(c)	→	(b)	(b)	(b)	(a)
(d)	→	(a)	(d)	(c)	(b)